

VANGSNES, Norway

WMO: 013380

Lat: 61.171N

Lon: 6.647E

Elev: 51

StdP: 100.71

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.6	-5.4	-13.5	1.2	-3.8	-12.0	1.3	-3.5	11.3	3.3	10.4	3.2	4.2	80	0.623

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.7	24.1	16.5	22.0	15.4	20.2	14.7	17.4	22.3	16.5	20.7	15.7	19.3	3.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.4	11.0	19.3	14.5	10.4	18.6	13.8	9.9	17.8	49.1	22.3	46.4	20.8	44.0	19.3	21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.2	8.2	7.3	-7.1	26.5	1.8	2.6	-8.4	28.4	-9.5	29.9	-10.6	31.3	-11.9	33.2		
			-8.5	18.3	1.8	1.2	-9.8	19.2	-10.8	20.0	-11.9	20.7	-13.2	21.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.9	1.1	1.2	2.9	6.6	10.7	13.9	16.3	15.2	11.9	7.8	4.2	2.1		
	DBStd	6.13	3.03	3.15	2.82	2.85	3.40	2.63	2.91	2.33	2.44	2.72	3.02	3.34		
	HDD10.0	1390	277	248	219	108	31	2	0	0	9	77	174	244		
	HDD18.3	3842	536	481	478	351	238	136	78	101	192	326	423	503		
	CDD10.0	613	0	0	0	6	53	119	196	162	67	10	1	0		
	CDD18.3	23	0	0	0	0	1	3	16	4	0	0	0	0		
	CDH23.3	94	0	0	0	0	6	12	69	7	0	0	0	0		
	CDH26.7	10	0	0	0	0	1	0	9	0	0	0	0	0		
	WSAvg	3.6	4.4	3.9	3.8	3.2	3.4	3.6	3.5	3.2	3.2	3.5	3.7	4.0		
Precipitation	PrecAvg	1304	149	128	120	67	55	68	81	88	120	139	134	159		
	PrecMax	1969	353	308	464	137	127	169	135	168	226	385	296	345		
	PrecMin	676	0	0	0	15	9	10	31	23	26	31	4	3		
	PrecStd	282	93	88	97	32	31	41	26	35	61	83	81	99		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.5	8.7	9.9	16.4	23.2	25.0	27.9	24.4	19.0	16.9	12.0	9.5		
		MCWB	5.9	5.1	5.4	9.5	14.4	15.4	18.3	16.8	13.5	11.7	8.6	6.6		
	2%	DB	7.2	7.4	8.5	14.4	20.2	22.3	25.5	21.7	17.5	14.1	10.6	8.3		
		MCWB	4.7	4.6	5.4	8.7	12.9	14.2	17.5	15.8	12.9	10.6	7.7	5.7		
	5%	DB	5.9	6.3	7.5	12.7	18.0	20.1	22.9	20.0	16.4	12.4	9.4	7.1		
		MCWB	3.9	3.7	4.8	7.6	11.8	13.0	16.5	15.3	12.4	9.5	6.9	4.9		
	10%	DB	4.8	5.2	6.6	10.9	15.9	18.2	21.0	18.8	15.4	11.2	8.2	6.1		
		MCWB	3.0	3.0	4.0	6.5	10.7	12.4	15.8	14.8	12.1	8.4	6.2	4.0		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.0	6.4	7.1	10.6	15.6	16.2	19.3	17.8	14.9	12.4	9.5	7.7		
		MCDB	7.9	7.8	8.5	15.2	22.1	23.2	25.4	22.5	17.2	14.7	10.9	8.8		
	2%	WB	5.2	5.1	6.2	9.2	13.5	14.8	18.2	16.8	14.0	11.2	8.5	6.1		
		MCDB	6.6	6.8	7.8	13.4	18.9	20.3	23.5	20.4	15.9	13.7	10.0	7.7		
	5%	WB	4.2	4.2	5.1	8.1	12.2	14.0	17.1	16.1	13.2	10.1	7.2	5.2		
		MCDB	5.6	5.6	7.0	11.6	16.8	18.3	21.9	19.1	15.3	12.2	8.8	6.6		
	10%	WB	3.3	3.4	4.4	7.1	11.1	13.3	16.2	15.4	12.6	9.0	6.1	4.4		
		MCDB	4.6	4.9	6.3	10.1	15.5	17.0	20.5	18.1	14.7	10.6	7.8	5.8		
Mean Daily Temperature Range	5% DB	MDBR	2.9	3.0	3.9	5.3	6.2	6.1	5.7	4.8	4.1	3.4	2.8	2.9		
		MCDBR	4.5	4.0	4.6	7.5	9.5	9.7	9.1	6.7	5.2	4.6	4.6	4.3		
	5% WB	MCWBR	3.4	2.7	3.0	3.7	4.3	4.2	3.6	3.1	2.8	2.9	3.2	3.3		
		MCWBR	4.1	4.0	4.4	6.6	8.7	7.7	8.2	6.0	4.4	4.3	4.4	4.2		
Clear-Sky Solar Irradiance	taub	taub	0.243	0.288	0.322	0.357	0.361	0.358	0.371	0.364	0.336	0.313	0.257	0.214		
		taud	2.077	2.188	2.160	2.141	2.293	2.402	2.398	2.416	2.482	2.414	2.153	1.978		
	Ebn at Noon	Ebn at Noon	477	685	784	826	859	870	846	822	784	665	455	326		
		Edh at Noon	39	68	99	123	115	105	103	94	74	56	36	25		
All-Sky Solar Radiation	RadAvg	RadAvg	0.22	0.73	1.94	3.69	4.79	4.85	4.61	3.53	2.20	1.01	0.32	0.14		
	RadStd	RadStd	0.05	0.21	0.40	0.63	0.73	0.53	0.52	0.35	0.28	0.15	0.05	0.02		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page